

An Analysis of Feature selection

Feature selection

Feature selection -- Part 1

Filter type methods select variables regardless of the model. They are based only on general features like the correlation with the variable to predict. Filter methods suppress the least interesting variables. The other variables will be part of a classification or a regression model used to classify or to predict data. These methods are particularly effective in computation time and robust to overfitting. Filter methods tend to select redundant variables when they do not consider the relationships between variables. However, more elaborate features try to minimize this problem by removing variables highly correlated to each other, such as the Fast Correlation Based Filter (FCBF) algorithm.

Feature selection -- Part 2

Exhaustive Best first Simulated annealing Genetic algorithm Greedy forward selection Greedy backward elimination Particle swarm optimization Targeted projection pursuit Scatter search Variable neighborhood search Two popular filter metrics for classification problems are correlation and mutual information, although neither are true metrics or 'distance measures' in the mathematical sense, since they fail to obey the triangle inequality and thus do not compute any actual 'distance' – they should rather be regarded as 'scores'. These scores are computed between a candidate feature (or set of features) and the desired output category. There are, however, true metrics that are a simple function of the mutual information; see here. Other available filter metrics include:

Feature selection -- Part 3

Q P F S : $\min_{\mathbf{x}} \{ \alpha \mathbf{x}^T \mathbf{H} \mathbf{x} - \mathbf{x}^T \mathbf{F} \}$ s.t. $\sum_{i=1}^n x_i = 1, x_i \geq 0$
$$\left(\alpha \mathbf{x}^T \mathbf{H} \mathbf{x} - \mathbf{x}^T \mathbf{F} \right) \quad \text{subject to} \quad \sum_{i=1}^n x_i = 1, x_i \geq 0$$

Feature selection -- Part 4

where $\mathbf{F}_{n \times 1} = [I(f_1; c), \dots, I(f_n; c)]^T$ is the vector of feature relevancy assuming there are n features in total, $\mathbf{H}_{n \times n} = [I(f_i; f_j)]_{i,j=1 \dots n}$ is the matrix of feature pairwise redundancy, and $\mathbf{x}_{n \times 1}$ represents relative feature weights. QPFS is solved via quadratic programming. It is recently shown that QPFS is biased towards features with smaller entropy, due to its placement of the feature self redundancy term $I(f_i; f_i)$ on the diagonal of $\mathbf{H}$.

Feature selection -- Part 5

Here, $\overline{r_{cf}}$ is the average value of all feature-classification correlations, and $\overline{r_{ff}}$ is the average value of all feature-feature correlations. The CFS criterion is defined as follows:

Feature selection -- Part 6

simplification of models to make them easier to interpret, shorter training times, to avoid the curse of dimensionality, improve the compatibility of the data with a certain learning model class, to encode inherent symmetries present in the input space. The central premise when using feature selection is that data sometimes contains features that are redundant or irrelevant, and can thus be removed without incurring much loss of information. Redundancy and irrelevance are two distinct notions, since one relevant feature may be redundant in the presence of another relevant feature with which it is strongly correlated. Feature extraction creates new features from functions of the original features, whereas feature selection finds a subset of the features. Feature selection techniques are often used in domains where there are many features and comparatively few samples (data points).

Feature selection -- Part 7

$D(S, c) = \frac{1}{|S|} \sum_{f_i \in S} I(f_i; c)$
 $D(S, c) = \frac{1}{|S|} \sum_{f_i \in S} I(f_i; c)$. The redundancy of all features in the set S is the average value of all mutual information values between the feature f_i and the feature f_j :

Feature selection -- Part 8

Minimum-redundancy-maximum-relevance (mRMR) feature selection Peng et al. proposed a feature selection method that can use either mutual information, correlation, or distance/similarity scores to select features. The aim is to penalise a feature's relevancy by its redundancy in the presence of the other selected features. The relevance of a feature set S for the class c is defined by the average value of all mutual information values between the individual feature f_i and the class c as follows:

Feature selection -- Part 9

H S I C L a s s o : $\min_{\mathbf{x}} \frac{1}{2} \sum_{k,l=1}^n x_k x_l \text{HSIC}(f_k, f_l) - \sum_{k=1}^n x_k \text{HSIC}(f_k, c) + \lambda \sum_{k=1}^n x_k$, s.t. $x_1, \dots, x_n \geq 0$,
$$\text{HSIC}_{\text{Lasso}} = \min_{\mathbf{x}} \left\{ \frac{1}{2} \sum_{k,l=1}^n x_k x_l \text{HSIC}(f_k, f_l) - \sum_{k=1}^n x_k \text{HSIC}(f_k, c) + \lambda \sum_{k=1}^n x_k \right\}$$

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$$\sum_{k=1}^n x_k \{\text{HSIC}\}(f_k, c) + \lambda \|\mathbf{w}\|_1, \quad \mathbf{x} \succeq \mathbf{0}$$

Feature selection -- Part 10

Correlation feature selection The correlation feature selection (CFS) measure evaluates subsets of features on the basis of the following hypothesis: "Good feature subsets contain features highly correlated with the classification, yet uncorrelated to each other". The following equation gives the merit of a feature subset S consisting of k features:

Feature selection -- Part 11

where $Q_{ii} = I(f_i; c)$ and $Q_{ij} = (I(f_i; c | f_j) + I(f_j; c | f_i)) / 2, i \neq j$. An advantage of SPECCMI is that it can be solved simply via finding the dominant eigenvector of Q , thus is very scalable. SPECCMI also handles second-order feature interaction.

Feature selection -- Part 12

Wrapper methods use a predictive model to score feature subsets. Each new subset is used to train a model, which is tested on a hold-out set. Counting the number of mistakes made on that hold-out set (the error rate of the model) gives the score for that subset. As wrapper methods train a new model for each subset, they are very computationally intensive, but usually provide the best performing feature set for that particular type of model or typical problem. Filter methods use a proxy measure instead of the error rate to score a feature subset. This measure is chosen to be fast to compute, while still capturing the usefulness of the feature set. Common measures include the mutual information, the pointwise mutual information, Pearson product-moment correlation coefficient, Relief-based algorithms, and inter/intra class distance or the scores of significance tests for each class/feature combinations. Filters are usually less computationally intensive than wrappers, but they produce a feature set which is not tuned to a specific type of predictive model. This lack of tuning means a feature set from a filter is more general than the set from a wrapper, usually giving lower prediction performance than a wrapper. However the feature set doesn't contain the assumptions of a prediction model, and so is more useful for exposing the relationships between the features. Many filters provide a feature ranking rather than an explicit best feature subset, and the cut off point in the ranking is chosen via cross-validation. Filter methods have also been used as a preprocessing step for wrapper methods, allowing a wrapper to be used on larger problems. One other popular approach is the Recursive Feature Elimination algorithm, commonly used with

Support Vector Machines to repeatedly construct a model and remove features with low weights. Embedded methods are a catch-all group of techniques which perform feature selection as part of the model construction process. The exemplar of this approach is the LASSO method for constructing a linear model, which penalizes the regression coefficients with an L1 penalty, shrinking many of them to zero. Any features which have non-zero regression coefficients are 'selected' by the LASSO algorithm. Improvements to the LASSO include Bolasso which bootstraps samples; Elastic net regularization, which combines the L1 penalty of LASSO with the L2 penalty of ridge regression; and FeaLect which scores all the features based on combinatorial analysis of regression coefficients. AEFS further extends LASSO to nonlinear scenario with autoencoders. These approaches tend to be between filters and wrappers in terms of computational complexity. In traditional regression analysis, the most popular form of feature selection is stepwise regression, which is a wrapper technique. It is a greedy algorithm that adds the best feature (or deletes the worst feature) at each round. The main control issue is deciding when to stop the algorithm. In machine learning, this is typically done by cross-validation. In statistics, some criteria are optimized. This leads to the inherent problem of nesting. More robust methods have been explored, such as branch and bound and piecewise linear network.